

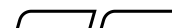
Aenderungsbeschreibung/Description of Revision war kZ		Kommt vor/Frequency
Buchst./Rev.Ltr. -.9	Aenderungs-Nr./Revision Notice No. PR063865	Bearbeitungsstatus/Lifecycle Released

Changes from Var1 Ed 01.02 to Var1 Ed 01.04
Corrections:
P-Exhaust Diff SCR1 Reg. 209 scaling changed from 100->10000 LM HI P-Exhaust Diff SCR Reg. 1198 scaling changed from 100->10000 LM HIHI P-Exhaust Diff SCR Reg. 1199 scaling changed from 100->10000 For binary register 0-10 FC 1 added -> read Analog Register 50-53 deleted
New measuring points added:
RA Concentration Register 299 P-Exhaust SCR Inlet1 Register 300 P-Exhaust SCR Inlet2 Register 301 P-Exhaust SCR Inlet 1 redundant Register 302 P-Exhaust Diff SCR2 Register 303 P-Dosing Air SUDU 1 Register 305 SCR System State1 Register 306 SCR System State2 Register 307 Actual RA Consumption Register 308 TripRAConsumption Register 309 TotalRAConsumption Register 311 Cmd Reset RA Consumption Register 800 MCS Analog added for GasMobile 250-294
New limits added:
LM HI P-Start Air Register 1101 LM HIHI P-Start Air Register 1102 LM LO RA Concentration Register 1103 LM LOLO RA Concentration Register 1104 LM HI RA Concentration Register 1105 LM HIHI RA Concentration Register 1106 LM HI P-Exhaust SCR Inlet1 Register 1107 LM HIHI P-Exhaust SCR Inlet1 Register 1108 LM HI P-Exhaust SCR Inlet2 Register 1109 LM HIHI P-Exhaust SCR Inlet2 Register 1110 LM HI P-Exhaust Diff SCR 2 Register 1111 LM HIHI P-Exhaust Diff SCR 2 Register 1112
New alarms added:
HI P-Start Air Register 1350 HIHI P-Start Air Register 1352 AL Wiring Emergency Stop SaSy Register 2456 AL Com. to NOxB Lost Register 4754 AL NOxB Warning Register 4756 AL NOxB Failure Register 4758 AL MaxAngleThrottleDeviation Register 4760 SD Engine Speed Crankshaft CPM Register 4762 SD Engine Speed Camshaft CPM Register 4764 AL EngineMisfireSeveralCylinder Register 4766 AL Press Sensor Warn A1 Register 4768 AL Press Sensor Warn A2 Register 4770

Tolerierung/Tolerancing ISO 8015		Masse/Size ISO 14405					
Allgemeintoleranzen/General Tolerances ISO 2768-mK							
Bauart/Type		Allgemeine Fertigungsvorschriften nach MMN332 Production Specification per MMN 332					
Genoline NewGeneration							
Ausf. u.Lieferung Tech. Characteristics							
Oberflaechenschutz Surface Protection		Oberflaechenangaben nach MTN5033 Surface Specification per MTN5033					
Verwendbar fuer Typ/Applicable to Model GLNG Release 20							
Projekt-/Auftrags-Nr./Project/Order No.							
Ref.-Nr./Ref. No. Type1 Var01 ED01_02		Emission ID					
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		Erstell. Drawn		2019-08-22 10:09:04		schaepers	
		Bearb. Change		2020-11-05 13:42:23		kellerma	
		Norm Std.				ihmor stef	
		Gepr. Checked				ihmor stef	
		Zeichnungs-Nr./Drawing No. ZNG00026576				Blatt/ Sheet	
		Beschreibung/Description Type1 Var01 ED01_04				1 von/of 44	

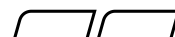
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AL_Press_Sensor_Warn_A3_Register 4772 AL_Press_Sensor_Warn_A4_Register 4774 AL_Press_Sensor_Warn_A5_Register 4776 AL_Press_Sensor_Warn_A6_Register 4778 AL_Press_Sensor_Warn_A7_Register 4780 AL_Press_Sensor_Warn_A8_Register 4782 AL_Press_Sensor_Warn_A9_Register 4784 AL_Press_Sensor_Warn_A10_Register 4786 AL_Press_Sensor_Warn_B1_Register 4788 AL_Press_Sensor_Warn_B2_Register 4790 AL_Press_Sensor_Warn_B3_Register 4792 AL_Press_Sensor_Warn_B4_Register 4794 AL_Press_Sensor_Warn_B5_Register 4796 AL_Press_Sensor_Warn_B6_Register 4798 AL_Press_Sensor_Warn_B7_Register 4800 AL_Press_Sensor_Warn_B8_Register 4802 AL_Press_Sensor_Warn_B9_Register 4804 AL_Press_Sensor_Warn_B10_Register 4806 SD_Engine_Speed_Crankshaft_IC_Register 4808 SD_Engine_Speed_Camshaft_IC_Register 4810 AL_Eng_Stop_Due_Close_Throttles_Register 4812 AL_P-Gas_Diff_Target_Actual_Register 4814 TD_Engine_Speed_ECU-CPM_Register 4816 HI_P-Start_Air_Register 4818 HIHI_P-Start_Air_Register 4820 AL_CPM_Sync_Speed_Sensors_Lost_Register 4822 SD_P_Gas_RailBankA_Register 4824 SD_P_Gas_RailBankB_Register 4826 SD_T_Gas_Register 4828 AL_IgnitionCtrlWarningA1_Register 4830 AL_IgnitionCtrlWarningA2_Register 4832 AL_IgnitionCtrlWarningA3_Register 4834 AL_IgnitionCtrlWarningA4_Register 4836 AL_IgnitionCtrlWarningA5_Register 4838 AL_IgnitionCtrlWarningA6_Register 4840 AL_IgnitionCtrlWarningA7_Register 4842 AL_IgnitionCtrlWarningA8_Register 4844 AL_IgnitionCtrlWarningA9_Register 4846 AL_IgnitionCtrlWarningA10_Register 4848 AL_IgnitionCtrlWarningB1_Register 4850 AL_IgnitionCtrlWarningB2_Register 4852 AL_IgnitionCtrlWarningB3_Register 4854 AL_IgnitionCtrlWarningB4_Register 4856 AL_IgnitionCtrlWarningB5_Register 4858 AL_IgnitionCtrlWarningB6_Register 4860 AL_IgnitionCtrlWarningB7_Register 4862 AL_IgnitionCtrlWarningB8_Register 4864 AL_IgnitionCtrlWarningB9_Register 4866 AL_IgnitionCtrlWarningB10_Register 4868 AL_IgnitionCtrlFailureA1_Register 4870 AL_IgnitionCtrlFailureA2_Register 4872 AL_IgnitionCtrlFailureA3_Register 4874 AL_IgnitionCtrlFailureA4_Register 4876 AL_IgnitionCtrlFailureA5_Register 4878 AL_IgnitionCtrlFailureA6_Register 4880 AL_IgnitionCtrlFailureA7_Register 4882 AL_IgnitionCtrlFailureA8_Register 4884 AL_IgnitionCtrlFailureA9_Register 4886 AL_IgnitionCtrlFailureA10_Register 4888 AL_IgnitionCtrlFailureB1_Register 4890 AL_IgnitionCtrlFailureB2_Register 4892 AL_IgnitionCtrlFailureB3_Register 4894 AL_IgnitionCtrlFailureB4_Register 4896 AL_IgnitionCtrlFailureB5_Register 4898 AL_IgnitionCtrlFailureB6_Register 4900 AL_IgnitionCtrlFailureB7_Register 4902 AL_IgnitionCtrlFailureB8_Register 4904 AL_IgnitionCtrlFailureB9_Register 4906 AL_IgnitionCtrlFailureB10_Register 4908 HI_T_IgnitionControl_Register 4910 HIHI_T_IgnitionControl_Register 4912 HI_T_IgnitionCtrlPowerOut_Register 4914 HIHI_T_IgnitionCtrlPowerOut_Register 4916 LO_T_Gas_GRU_Register 4918 LOLO_T_Gas_GRU_Register 4920 HI_T_Gas_GRU_Register 4922 HIHI_T_Gas_GRU_Register 4924 LO_P_Gas_GRUInlet_Register 4926 HI_P_Gas_GRUInlet_Register 4928 HI_GRU_PowerSupplyInlet_Register 4930 LO_GRU_PowerSupplyInlet_Register 4932 AL_IgnitionCtrlFailureComb_Register 4934 AL_IgnitionCtrlIntErr_Register 4936 AL_CylinderMonitoringFail_Register 4938 LO_P_CtrlAirGRUInlet_Register 4940 LOLO_P_CtrlAirGRUInlet_Register 4942 AL_UnburnedFuelInExhaust_Register 4944 AL_Emission_Compliance_Fail_Register 5000 SD_P-Exhaust_Diff_SCR1_Register 5002 AL_Com_Lost_NH3_SCR_Outlet1_Register 5004 SD_Ammonia_NH3_SCR_Outlet1_Register 5006 AL_EGAT_Bypass_Position_Fault_Register 5008 HI_P-Exhaust_SCR_Inlet1_Register 5010 HIHI_P-Exhaust_SCR_Inlet1_Register 5012 SD_P-Exhaust_SCR_Red_Inlet1_Register 5014 TD_P-Exhaust_SCR_Inlet1_Register 5016 HI_P-Exhaust_SCR_Inlet2_Register 5018 HIHI_P-Exhaust_SCR_Inlet2_Register 5020 LO_SXU_Power_Supply_Voltage_Register 5022 LOLO_SXU_Power_Supply_Voltage_Register 5024 HI_SXU_Power_Supply_Voltage_Register 5026 HIHI_SXU_Power_Supply_Voltage_Register 5028 HI_T-SXU_Register 5030 AL_SXU_Internal_Error_Register 5032 HI_SCR_HC1_Deposition_Register 5034 HIHI_SCR_HC1_Deposition_Register 5036 HI_SCR_HC2_Deposition_Register 5038 HIHI_SCR_HC2_Deposition_Register 5040		
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HI SCR RA Deposition Register 5042 HIHI SCR RA Deposition Register 5044 HI Ammonia NH3 SCR Outlet 1 Register 5046 HIHI Ammonia NH3 SCR Out 1 Register 5048 AL Com. Lost NH3 SCR Outlet 2 Register 5050 SD Ammonia NH3 SCR Outlet 2 Register 5052 HI Ammonia NH3 SCR Outlet 2 Register 5054 HIHI Ammonia NH3 SCR Out 2 Register 5056 LO T-Reducing Agent SUDU F1 Register 5058 LO P-Reducing Agent DU 1 Register 5060 HI P-Reducing Agent DU 1 Register 5062 HI Exhaust NOx Register 5064 AL ElectricalFault VND Register 5066 AL ElectricalFault VNC Register 5068 AL HydraulicsFault VND Register 5070 AL HydraulicsFault VNR Register 5072 AL RA Supply Unit 1 Defect Register 5074 AL RA Supply Unit 2 Defect Register 5076 AL RA Supply Unit 3 Defect Register 5078 AL RA Supply Unit 4 Defect Register 5080 AL RA Supply Unit 5 Defect Register 5082 AL RA Supply Unit 6 Defect Register 5084 SD P-Fuel bef. interm. Filter Register 5086 AL Injector Defect A1 Register 5088 AL Injector Defect A2 Register 5090 AL Injector Defect A3 Register 5092 AL Injector Defect A4 Register 5094 AL Injector Defect A5 Register 5096 AL Injector Defect A6 Register 5098 AL Injector Defect A7 Register 5100 AL Injector Defect A8 Register 5102 AL Injector Defect A9 Register 5104 AL Injector Defect A10 Register 5106 AL Injector Defect B1 Register 5108 AL Injector Defect B2 Register 5110 AL Injector Defect B3 Register 5112 AL Injector Defect B4 Register 5114 AL Injector Defect B5 Register 5116 AL Injector Defect B6 Register 5118 AL Injector Defect B7 Register 5120 AL Injector Defect B8 Register 5122 AL Injector Defect B9 Register 5124 AL Injector Defect B10 Register 5126 AL Injector Defect critical A1 Register 5128 AL Injector Defect critical A2 Register 5130 AL Injector Defect critical A3 Register 5132 AL Injector Defect critical A4 Register 5134 AL Injector Defect critical A5 Register 5136 AL Injector Defect critical A6 Register 5138 AL Injector Defect critical A7 Register 5140 AL Injector Defect critical A8 Register 5142 AL Injector Defect critical A9 Register 5144 AL Injector Defect critical A10 Register 5146 AL Injector Defect critical B1 Register 5148 AL Injector Defect critical B2 Register 5150 AL Injector Defect critical B3 Register 5152 AL Injector Defect critical B4 Register 5154 AL Injector Defect critical B5 Register 5156 AL Injector Defect critical B6 Register 5158 AL Injector Defect critical B7 Register 5160 AL Injector Defect critical B8 Register 5162 AL Injector Defect critical B9 Register 5164 AL Injector Defect critical B10 Register 5166 LO P-Reducing Agent PFM Register 5168 HI P-Reducing Agent PFM Register 5170 AL Injector Double Defect Register 5172 SD P-Exhaust Diff. SCR 2 Register 5174 SD P-Exhaust SCR Inlet 2 Register 5176 SD P-Exhaust SCR Outlet 2 Register 5178	
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Aenderungs-Nr./Revision Notice No.
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Bearbeitungsstatus/Lifecycle
Released

MCS Analog Measuring Points

Function Code	03 Holding Register
	16 Write Holding Register
Minimum Cycle Time	200ms
Missing Data:	0x7FFF
Sensor Defect	0x8001
Sensor Defect Combined Registers (32Bit)	0x80000001

Register	Denomination	Min	Max	Unit	Scaling	Read/Write
0	Engine Speed	0	3000	rpm	10	R
1	Engine Speed SaSy	0	3000	rpm	10	R
2	Engine Speed Camshaft	0	3000	rpm	10	R
3	Engine Speed Crankshaft	0	3000	rpm	10	R
4	Idle Speed	0	3000	rpm	10	R
5	Nominal Speed	0	3000	rpm	10	R
6	Feedback Speed Demand	0	3000	rpm	10	R
7	Feedback Speed Demand Eff.	0	3000	rpm	10	R
11	ETC 1 Speed	0	100	krpm	100	R
12	ETC 2 Speed	0	100	krpm	100	R
13	ETC 3 Speed	0	100	krpm	100	R
14	ETC 4 Speed	0	100	krpm	100	R
24	Power Supply EIM/Starter Bat.	0	50	V	100	R
25	ECU Power Supply Voltage	0	35	V	100	R
26	U-SPU Main Power Supply	0	50	V	100	R
27	U-SPU Emerg. Power Supply	0	50	V	100	R
30	P-Lube Oil aft. Filter (ECU)	0	10	bar	100	R
31	P-Lube Oil aft. Filter SaSy	0	10	bar	100	R
32	P-Lube Oil A-Side ETCs	0	120	bar	100	R
33	P-Lube Oil B-Side ETCs	0	120	bar	100	R
34	P-Oil after Level Pump	0	10	bar	100	R
35	P-Lube Oil last bearing	0	10	bar	100	R
36	P-Piston Cooling Oil	0	6	bar	100	R
37	P-Piston Cool.Oil Filter Diff.	0	10	bar	100	R
38	P-Lube Oil before Filter (ECU)	0	10	bar	100	R
39	P-Oil Filter Difference	0	16	bar	100	R
42	P-Oil Refill Pump	0	6	bar	100	R
43	P-Coolant	0	6	bar	100	R
44	P-Raw Water before Pump	0	6	bar	100	R
45	P-Raw Water	0	6	bar	100	R
46	P-Fuel	0	15	bar	100	R
47	P-Fuel before Filter (ECU)	0	10	bar	100	R
48	P-Fuel Prefilter Diff.	0	10	bar	100	R
49	P-Fuel Filter Difference	0	16	bar	100	R
54	P-Crankcase	-70	70	mbar	100	R
55	P-Intake Air B	-70	70	mbar	100	R
56	P-Charge Air A	0	6	bar	100	R
57	P-Charge Air B	0	6	bar	100	R
58	P-Control Air	0	10	bar	100	R
59	P-Start Air	0	50	bar	100	R
60	P-Exhaust back	0	200	bar	100	R
62	UTC Seconds	0	59	digit	1	R
63	UTC Minutes	0	59	digit	1	R
64	UTC Hour	0	23	digit	1	R
65	Month	1	12	digit	1	R
66	Day	1	31	digit	1	R
67	Year	1970	2060	digit	1	R
70	T-Lube Oil	-20	120	degC	10	R
71	T-Lube Oil ETC	-20	120	degC	10	R
75	T-Fuel bef.Low Press Pump	-20	120	degC	10	R
76	T-Fuel Common Rail A	-20	120	degC	10	R
77	T-Fuel Common Rail B	-20	120	degC	10	R
78	T-Coolant (ECU)	-20	120	degC	10	R
79	T-Coolant SaSy	-20	120	degC	10	R
80	T-Raw Water	-20	120	degC	10	R
81	T-Intake Air	-20	120	degC	10	R
82	T-Charge Air A	-20	120	degC	10	R
83	T-Charge Air B	-20	120	degC	10	R
84	T-Charge Air Seq Ctrl Valve	0	1000	degC	10	R
85	T-Exhaust Combined A	-20	800	degC	10	R
86	T-Exhaust before ETC (Combined B)	-20	800	degC	10	R



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Aenderungsbeschreibung/Description of Revision
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Bearbeitungsstatus/Lifecycle
Released

Register	Denomination	Min	Max	Unit	Scaling	Read/Write
87	T-Exhaust Combined C	-20	1000	degC	10	R
88	T-Exhaust Combined D	-20	1000	degC	10	R
89	T-Exhaust A1	0	1000	degC	10	R
90	T-Exhaust A2	0	1000	degC	10	R
91	T-Exhaust A3	0	1000	degC	10	R
92	T-Exhaust A4	0	1000	degC	10	R
93	T-Exhaust A5	0	1000	degC	10	R
94	T-Exhaust A6	0	1000	degC	10	R
95	T-Exhaust A7	0	1000	degC	10	R
96	T-Exhaust A8	0	1000	degC	10	R
97	T-Exhaust A9	0	1000	degC	10	R
98	T-Exhaust A10	0	1000	degC	10	R
99	T-Exhaust B1	0	1000	degC	10	R
100	T-Exhaust B2	0	1000	degC	10	R
101	T-Exhaust B3	0	1000	degC	10	R
102	T-Exhaust B4	0	1000	degC	10	R
103	T-Exhaust B5	0	1000	degC	10	R
104	T-Exhaust B6	0	1000	degC	10	R
105	T-Exhaust B7	0	1000	degC	10	R
106	T-Exhaust B8	0	1000	degC	10	R
107	T-Exhaust B9	0	1000	degC	10	R
108	T-Exhaust B10	0	1000	degC	10	R
109	T-Exhaust Mean	0	1000	degC	10	R
110	T-Bearing 1	0	200	degC	10	R
111	T-Bearing 2	0	200	degC	10	R
112	T-Bearing 3	0	200	degC	10	R
113	T-Bearing 4	0	200	degC	10	R
114	T-Bearing 5	0	200	degC	10	R
115	T-Bearing 6	0	200	degC	10	R
116	T-Bearing 7	0	200	degC	10	R
117	T-Bearing 8	0	200	degC	10	R
118	T-Bearing 9	0	200	degC	10	R
119	T-Bearing 10	0	200	degC	10	R
120	T-Bearing 11	0	200	degC	10	R
121	T-Bearing Mean	0	200	degC	10	R
122	T-Splash Oil B1	-40	150	degC	10	R
123	T-Splash Oil B2	-40	150	degC	10	R
124	T-Splash Oil B3	-40	150	degC	10	R
125	T-Splash Oil B4	-40	150	degC	10	R
126	T-Splash Oil B5	-40	150	degC	10	R
127	T-Splash Oil B6	-40	150	degC	10	R
128	T-Splash Oil B7	-40	150	degC	10	R
129	T-Splash Oil B8	-40	150	degC	10	R
130	T-Splash Oil B9	-40	150	degC	10	R
131	T-Splash Oil B10	-40	150	degC	10	R
132	T-Splash Oil Mean	-40	150	degC	10	R
133	T-ECU	0	100	degC	10	R
135	T-Exhaust after Engine	0	400	degC	10	R
140	ECU Failure Codes	0	5000	digit	1	R
141	Torque Limitation Code	-2147483646	2147483646	digit	1	R
142	SPU Interface Version HI	-2147483646	2147483646	digit	1	R
143	SPU Interface Version LO					
144	MCS Version HI	-2147483646	2147483646	digit	1	R
145	MCS Version LO					
146	SPU Alarm Groups coded static HI	-2147483646	2147483646	digit	1	R
147	SPU Alarm Groups coded static LO					
148	SPU Alarm Groups coded dyn. HI					
149	SPU Alarm Groups coded dyn. LO	-2147483646	2147483646	digit	1	R
150	Alarm Counter	0	65535	digit	1	R
152	Status Start Procedure HI	-2147483646	2147483646	digit	1	R
153	Status Start Procedure LO					
155	Trip Fuel Consumption HI	0	200000000	l	1	R
156	Trip Fuel Consumption LO					
157	Total Fuel Consumption HI					
158	Total Fuel Consumption LO	0	200000000	l	1	R
159	Actual Fuel Consumption	0	3000	l/h	10	R
160	Mean Trip Fuel Consumption	0	3000	l/h	10	R
161	Fuel Consumption Max	0	3000	l/h	10	R
162	Fuel Consumption Act	0	100	%	100	R
163	Fuel Consumption Mean Trip	0	100	%	100	R
165	Injection Limitation 0-120%	0	120	%	100	R
166	Injection in Relation to DBR	0	120	%	100	R



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Register	Denomination	Min	Max	Unit	Scaling	Read/Write
170	ECU Operat.Hours HI	0	100000	h	1	R
171	ECU Operat.Hours LO					
172	Trip Duration HI					
173	Trip Duration LO	0	100000	h	1	R
176	Engine Load Reserve					
178	Injection Quantity 0-120					
183	User Instruction	0	65535	digit	1	R
204	Cylinder Cutout Mask HI	-2147483646	2147483646	digit	1	R
205	Cylinder Cutout Mask LO					
206	T-Exhaust SCR Inlet1					
207	T-Exhaust SCR Inlet2	0	1000	degC	10	R
208	T-Exhaust SCR Outlet1	0	1000	degC	10	R
209	P-Exhaust Diff SCR1	0	10	bar	100	R
210	P-RA Filter Out At PFM	0	10	bar	100	R
211	T-Reducing Agent At PFM	0	1000	degC	10	R
212	RA Tank Level Norm	0	100	%	100	R
213	P-Fuel Prefilter	0	10	bar	100	R
220	T-Exhaust SCR Outlet 2	0	1000	degC	10	R
221	Power Actual Gen	0	2000	kW	10	R
231	T-Bearing GenDE	-50	200	degC	10	R
232	T-Bearing GenNDE	-50	200	degC	10	R
233	T-AirInlet Gen	-50	200	degC	10	R
234	T-AirOutlet Gen	-50	200	degC	10	R
236	T-WindingL1 Gen	-50	200	degC	10	R
237	T-WindingL2 Gen	-50	200	degC	10	R
238	T-WindingL3 Gen	-50	200	degC	10	R
239	T-WindingL4 Gen	-50	200	degC	10	R
246	T-Coolant Gen Out	-50	200	degC	10	R
247	T-Coolant Gen In	-50	200	degC	10	R
250	CPM Sensor Configuration 1	0	2147483646	digit	1	R
251	Cylinder Performance Cyl. A1	0	100	%	100	R
252	Cylinder Performance Cyl. A2	0	100	%	100	R
253	Cylinder Performance Cyl. A3	0	100	%	100	R
254	Cylinder Performance Cyl. A4	0	100	%	100	R
255	Cylinder Performance Cyl. A5	0	100	%	100	R
256	Cylinder Performance Cyl. A6	0	100	%	100	R
257	Cylinder Performance Cyl. A7	0	100	%	100	R
258	Cylinder Performance Cyl. A8	0	100	%	100	R
259	Cylinder Performance Cyl. A9	0	100	%	100	R
260	Cylinder Performance Cyl. A10	0	100	%	100	R
261	Cylinder Performance Cyl. B1	0	100	%	100	R
262	Cylinder Performance Cyl. B2	0	100	%	100	R
263	Cylinder Performance Cyl. B3	0	100	%	100	R
264	Cylinder Performance Cyl. B4	0	100	%	100	R
265	Cylinder Performance Cyl. B5	0	100	%	100	R
266	Cylinder Performance Cyl. B6	0	100	%	100	R
267	Cylinder Performance Cyl. B7	0	100	%	100	R
268	Cylinder Performance Cyl. B8	0	100	%	100	R
269	Cylinder Performance Cyl. B9	0	100	%	100	R
270	Cylinder Performance Cyl. B10	0	100	%	100	R
271	Cylinder Performance Mean	0	100	%	100	R
272	P-Gas A	0	1000	bar	100	R
273	T-Gas	0	1000	degC	10	R
274	Actual Gas Consumption	0	214748364	Kg/h	10	R
275	Trip Gas Consumption	0	214748364	t	10	R
276	Total Gas Consumption	0	214748364	t	10	R
277	GRU State	0	2147483646	digit	1	R
278	P-Gas Inlet	0	1000	bar	100	R
279	P-Gas Outlet	0	1000	bar	100	R
280	T-Gas GRU	0	1000	degC	10	R
281	P-Gas Set Point	0	1000	bar	100	R
282	Throttle Flap A Actual Value	0	100	%	100	R
283	Throttle Flap B Actual Value	0	100	%	100	R
284	P-Gas B	0	1000	bar	100	R
285	Mean Trip Gas Consumption	0	214748364	Kg/h	10	R
286	Max. Scale Gas Consumption	0	214748364	Kg/h	10	R
287	Actual Gas Cons. Bargraph	0	100	%	100	R
288	Mean Trip Gas Cons. Bargraph	0	100	%	100	R
289	T-EMU	0	100	degC	10	R
290	EMU Power Supply Voltage	0	50	V	10	R
291	T-CPM	0	100	degC	10	R
292	CPM Power Supply Voltage	0	50	V	10	R



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Register	Denomination	Min	Max	Unit	Scaling	Read/Write
293	T-Coolant Intercooler	0	1000	degC	10	R
294	T-EIM	0	100	degC	10	R
295	MMS Errorcodes	0	2147483646	digit	1	R
299	RA Concentration	0	100	%	100	R
300	P-Exhaust SCR Inlet1	0	2	bar	10000	R
301	P-Exhaust SCR Inlet2	0	2	bar	10000	R
302	P-Exhaust SCR Inlet 1 redundant	0	2	bar	10000	R
303	P-Exhaust Diff SCR2	0	0,2	bar	10000	R
305	P-Dosing Air SUDU 1	0	10	bar	100	R
306	SCR System State1	0	2147483646	digit	1	R
307	SCR System State2	0	2147483646	digit	1	R
308	Actual RA Consumption	0	100	l/h	10	R
309	TripRAConsumption HI	0	200000000	l	1	R
310	TripRAConsumption LO					
311	TotalRAConsumption HI	0	200000000	l	1	R
312	TotalRAConsumption LO					
500	Adjust UTC Seconds	0	59	digit	1	W
501	Adjust UTC Minutes	0	59	digit	1	W
502	Adjust UTC Hours	0	23	digit	1	W
503	Adjust Day	1	31	digit	1	W
504	Adjust Month	1	12	digit	1	W
505	Adjust Year	1975	2060	digit	1	W



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MCS Alarms

Function Code		01 Read Coils
Minimum Cycle Time for synchronisation and initialisation		5000ms
Analog-Register 150 (Alarm Counter) is being changed if new Alarm occurs or Alarm disappears. If Register-Value has changed since last request, read all Alarm-Informationen at once.		
2 Bits per Alarm		
High-Bit	Low-Bit	
0	0	No Alarm
0	1	Acknowledged Alarm
1	0	Spare/not used
1	1	Unacknowledged Alarm

Register	Denomination	Alarm Priority	Alarm Description
1002	LOLO Engine Speed	Yellow (3)	Engine is being stalled. The engine speed of the normally operating engine dropped below the limit from parameter 2.2500.027 Limit Engine Speed Low without any stop request. For safety reason the engine is stopped when this event occurs.
1004	HIHI Engine Overspeed (EIM)	Red (2)	Highest Limit of engine overspeed
1006	HIHI Engine Overspeed SaSy	Red (2)	An engine overspeed was detected by the safety system. This alarm will normally cause an automatic emergency stop.
1008	SD Engine Speed Camshaft	Yellow (4)	Sensor defect on camshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B1), if necessary replace it. After Engine restart follows curing of failure.
1010	SD Engine Speed Crankshaft	Yellow (4)	Sensor defect on crankshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B13), if necessary replace it. After Engine restart follows curing of failure.
1018	SD ETC 1 Speed	Yellow (4)	Speed-sensor of basic charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.1), if necessary replace it.
1020	HIHI ETC 1 Overspeed	Red (2)	Speed of basic charger too high (limit 2).
1022	HI ETC 1 Speed	Yellow (3)	Speed of basic charger too high (limit 1).
1024	SD ETC 2 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.
1026	HIHI ETC 2 Overspeed	Red (2)	Speed of 1st switchable charger too high (limit 2).
1028	HI ETC 2 Speed	Yellow (3)	Speed of 1st switchable charger too high (limit 1).
1030	SD ETC 3 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.
1032	HIHI ETC 3 Overspeed	Red (2)	Speed of 2nd switchable charger too high (limit 2).
1034	HI ETC 3 Speed	Yellow (3)	Speed of 2nd switchable charger too high (limit 1).
1036	SD ETC 4 Speed	Yellow (4)	Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.4), if necessary replace it.
1038	HIHI ETC 4 Overspeed	Red (2)	Speed of 3rd switchable charger too high (limit 2).
1040	HI ETC 4 Speed	Yellow (3)	Speed of 3rd switchable charger too high (limit 1).
1048	HI ETC Speed Difference	Yellow (4)	Speed deviation between basic turbo charger and one of the switchable chargers.
1110	LO Starter Supply Voltage	Yellow (4)	The starter supply voltage is below the minimum limit. The starter supply voltage is not monitored during the start procedure.
1112	SD ECU Power Supply Voltage	Yellow (4)	Internal ECU error. ==> electronic defect.
1114	HIHI ECU Power Supply Voltage	Red (2)	Power supply voltage too high (limit 2) ==> Check battery / generator
1116	HI ECU Power Supply Voltage	Yellow (4)	Power supply voltage too high (limit 1) ==> Check battery / generator
1118	LO ECU Power Supply Voltage	Yellow (4)	Power supply voltage too low (limit 1) ==> Check battery / generator
1120	LOLO ECU Power Supply Voltage	Red (2)	Power supply voltage too low (limit 2) ==> Check battery / generator
1122	SD U-Power Driver Unit	Yellow (4)	Sensor defect of Injector Power driver unit. ==> Internal error of ECU7. Change ECU7.
1124	HIHI U-Power Driver Unit	Yellow (3)	Power driver voltage (injectors) too high (limit 2).
1126	HI U-Power Driver Unit	Yellow (3)	Power driver voltage (injectors) too high (limit 1).
1128	LO U-Power Driver Unit	Yellow (4)	Power driver voltage (injectors) too low (limit 1).
1130	LOLO U-Power Driver Unit	Red (2)	Power driver voltage (injectors) too low (limit 2).
1132	HI U-SPU Main Power Supply	Yellow (3)	The voltage of the SPU main power supply is too high.
1134	LO U-SPU Main Power Supply	Yellow (3)	The voltage of the SPU main power supply is too low.



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Register	Denomination	Alarm Priority	Alarm Description
1136	HI U-SPU Emerg.Power Supply	Yellow (3)	The voltage of the SPU emergency power supply is too high.
1138	LO U-SPU Emerg.Power Supply	Yellow (3)	The voltage of the SPU emergency power supply is too low.
1140	LO U-SPU Real Time Clock	Yellow (3)	The voltage of backup battery for time and date in the LOP is too low, logging features may have an incorrect time stamp. Replace the battery located in the SPU/LOP!
1142	AL EMU 11V	Yellow (4)	The internal 11V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.
1144	AL EMU 7V	Yellow (4)	The internal -7V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.
1146	AL U-SPU Main Power SupplyFail	Yellow (4)	The main power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.
1148	AL U-SPU Emerg. Pow Sup Fail	Yellow (4)	The emergency power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.
1150	SD P-Lube Oil	Yellow (4)	Lube oil pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.
1152	LO P-Lube Oil	Yellow (3)	Pressure of lube oil too low (limit 1).
1154	LOLO P-Lube Oil	Red (2)	Pressure of lube oil too low (limit 2).
1156	SD P-Lube Oil SaSy	Yellow (4)	The safety system sensor signal for lube oil pressure is outside the defined sensor signal range. The safety system lube oil pressure information is not available anymore for indication or monitoring. The safety functions related to this signal are not available anymore.
1158	LO P-Lube Oil SaSy	Yellow (3)	A low engine lube oil pressure was detected by the safety system. A worsening situation could trigger a safety reaction.
1160	LOLO P-Lube Oil SaSy	Red (2)	A very low engine lube oil pressure was detected by the safety system. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.
1162	TD P-Lube Oil	Yellow (4)	The pressure difference between the redundant lube oil pressure sensors is too high.
1164	SD P-Lube Oil A-Side ETCs	Yellow (4)	Lube oil pressure-sensor of ETC on engine A-Side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.
1166	LO P-Lube Oil A-Side ETCs	Yellow (3)	A low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.
1168	LOLO P-Lube Oil A-Side ETCs	Red (2)	A very low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with a separate alarm.
1170	SD P-Lube Oil B-Side ETCs	Yellow (4)	Lube oil pressure-sensor of ETC on engine B-side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.
1172	LO P-Lube Oil B-Side ETCs	Yellow (3)	A low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.
1174	LOLO P-Lube Oil B-Side ETCs	Red (2)	A very low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.
1176	SD P-Oil after Level Pump	Yellow (4)	Sensor defect of lube oil pressure after level pump
1178	LO P-Oil after Level Pump	Yellow (3)	Low Limit of lube oil pressure after level pump
1180	SD P-Oil bef.High Press.Pump A	Yellow (4)	Sensor defect of lube oil pressure before high pressure pump in A
1182	LO P-Oil bef.High Press.Pump A	Yellow (3)	Low Limit of lube oil pressure before high pressure pump A
1184	SD P-Oil bef.High Press.Pump B	Yellow (4)	Sensor defect of lube oil pressure before high pressure pump in B
1186	LO P-Oil bef.High Press.Pump B	Yellow (3)	Low Limit of lube oil pressure before high pressure pump B
1204	SD P-Lube Oil before Filter	Yellow (4)	Pressure sensor for lube oil before filter defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.
1206	SD P-Oil Filter Difference	Yellow (4)	Sensor for differential pressure of lube oil defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.
1208	HIHI P-Oil Filter Difference	Red (2)	Differential pressure of oil filter too high (limit 2).
1210	HI P-Oil Filter Difference	Yellow (3)	Differential pressure of oil filter too high (limit 1).
1226	SD P-Oil Refill Pump	Yellow (4)	The pressure sensor signal for the oil refill pump pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.
1228	LO P-Oil Refill Pump	Yellow (3)	The oil pressure after the oil refill pump which is pumping oil out and back into the oil pan was low for a certain period (several minutes). As long as the pump is able to pump oil out and back into the oil pan, there is enough oil in the engine. If the oil level is below the end of the "pump snorkel" the pump will not produce pressure within a certain time and this alarm will be triggered.
1232	SD P-Coolant	Yellow (4)	Coolant pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B16), if necessary replace it.
1234	LO P-Coolant	Yellow (3)	Coolant pressure too low (limit 1) ==> check cooling cycle.
1236	LOLO P-Coolant	Red (2)	Coolant pressure too low (limit 2) ==> Engine stop or limitation of the injection quantity ==> check cooling cycle
1242	SD P-Raw Water	Yellow (4)	The pressure sensor signal for the raw water pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.
1244	LO P-Raw Water	Yellow (3)	The raw water pressure after the pump is too low
1246	SD P-Fuel after Filter	Yellow (4)	Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B34), if necessary replace it.
1248	LO P-Fuel	Yellow (3)	Fuel supply pressure too low (limit 1) ==> Check filter, fuel on low-pressure side



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1250	LOLO P-Fuel	Red (2)	Fuel supply pressure too low (limit 2) ==> Check filter (low-pressure side)
1252	SD P-Fuel before Filter	Yellow (4)	Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.
1254	SD Fuel Prefilter 1	Yellow (4)	The self test function of the fuel prefilter sensor 1 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).
1256	SD Fuel Prefilter 2	Yellow (4)	The self test function of the fuel prefilter sensor 2 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).
1258	HI P-Fuel Int. Filter Diff.	Yellow (3)	Fuel Prefilter Difference to high
1260	SD P-Fuel Filter Difference	Yellow (4)	Sensor for differential pressure of fuel defect. ==> only if SD-Alarms Fuel b. Filter or Fuel after Filter
1262	HI P-Fuel Filter Difference	Yellow (3)	Differential pressure of fuel filter too high (limit 1).
1264	SD P-Fuel bef. Low Press. Pump	Yellow (4)	Sensor defect of fuel pressure before low pressure pump
1266	LO P-Fuel bef. Low Press. Pump	Yellow (3)	Low Limit of fuel pressure before low pressure pump
1268	SD P-Fuel Common Rail	Yellow (4)	Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.
1270	HI P-Fuel Common Rail	Yellow (3)	Rail pressure > set point value ==> DBR reduction, shift of start of injection delayed (==> interphase transformer sticks or connections of the interphase transformer)
1272	LOLO P-Fuel Common Rail	Red (2)	Rail pressure < setpoint value ==> DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1274	LO P-Fuel Common Rail	Yellow (3)	Rail pressure < set point value ==> DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1276	SD P-Fuel Common Rail B	Yellow (4)	Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.
1278	HI P-Fuel Common Rail B	Yellow (3)	High Limit of fuel pressure in rail B
1280	LOLO P-Fuel Common Rail B	Red (2)	Rail pressure on Rail B < setpoint value ==> DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1282	LO P-Fuel Common Rail B	Yellow (3)	Rail pressure < setpoint value ==> DBR reduction (==> interphase transformer defective or leakage in high-pressure system)
1296	SD P-Fuel Return Path	Yellow (4)	Sensor defect of fuel pressure in return path
1298	HI P-Fuel Return Path	Yellow (3)	High Limit of fuel pressure in return path
1300	SD P-Crankcase	Yellow (4)	Crankcase pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B50), if necessary replace it.
1302	HIHI P-Crankcase	Red (2)	Crankcase pressure too high (limit 2).
1304	HI P-Crankcase	Yellow (3)	Crankcase pressure too high (limit 1).
1306	LO P-Intake Air After FilterB	Yellow (3)	Low Limit of intake air pressure B reached
1308	SD P-Charge Air A	Yellow (4)	Charge air pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10), if necessary replace it.
1310	HIHI P-Charge Air A	Red (2)	Pressure of charge air too high (limit 2).
1312	HI P-Charge Air A	Yellow (3)	Pressure of charge air too high (limit 1).
1314	SD P-Charge Air B	Yellow (4)	Charge Air Pressur B-Side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10.11), if necessary replace it. After Engine restart follows curing of failure.
1316	HIHI P-Charge Air B	Red (2)	Highest Limit of charge air pressure in B reached
1318	HI P-Charge Air B	Yellow (3)	High Limit of charge air pressure in B reached
1324	SD P-Start Air	Yellow (4)	The pressure sensor signal for the start air pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal (start interlock if start air pressure is too low) are not available anymore.
1326	LO P-Start Air	Yellow (4)	The pressure for the air starter is too low.
1328	LOLO P-Start Air	Yellow (3)	The pressure for the air starter is very low and a start is interlocked. The limit for that alarm should be set in a way that air for at least one start is remaining. This is to prevent emptying the air supply by (automatic) repeated starts. The start interlock can be disabled by activating "Override"
1350	HI P-Start Air	Yellow (3)	The pressure for the air starter is too high. A high air pressure can damage the air starter.
1352	HIHI P-Start Air	Red (2)	The pressure for the air starter is too high for the air starter and a start is interlocked to protect the air starter against damage. The start interlock can be disabled by activating "Override"
1386	SD T-Lube Oil	Yellow (4)	Lube oil temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B7), if necessary replace it.
1388	HIHI T-Lube Oil	Red (2)	Lube oil temperature to high (limit 2).
1390	HI T-Lube Oil	Yellow (3)	Lube oil temperature too high (limit 1).
1394	SD T-Lube Oil ETCs	Yellow (4)	T-Lube Oil turbo charger sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable, if necessary replace it.
1396	HIHI T-Lube Oil ETCs	Red (2)	Lube oil temperature of the basic charger is too high (above second limit).



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